CS 532, Spring 2020: Quiz 2

NAME:

Problem 1. Let $\mathbf{A} = (X, Y, Z)$, denote a 3D point expressed in the camera coordinate system. Let $\mathbf{a}$ be the intersection between point A's viewing ray and the image plane, what are the 3D coordinates of $\mathbf{a}$ If the camera focal length is $f = Z/10$. Make sketch.

Problem 2. Develop the expression for the cross product between $\mathbf{A}$ and $\mathbf{a}$